Detroit 2023 - 2025 Model I

Elliot Dean
Office of the Chief Financial Officer | Office of the Assessor
City of Detroit

6 August 2025

Contents

Load Necessary Libraries

```
library(tidyverse)   # data wrangling & ggplot2
library(janitor)     # functions for examining and cleaning dirty data
library(haven)       # to read SPSS .sav files
library(lmtest)      # Breusch-Pagan test
library(ggpubr)      # Automatically print the regression equation for ggplot
library(spgwr)       # Functions for computing geographically weighted regressions
library(tidymodels)  # collection of packages for modeling and machine learning
library(xgboost)     # Extreme Gradient Boosting
library(bundle)      # Serialize Model Objects with a Consistent Interface
library(psych)       # group statistics, including correlations and factor analysis
library(kableExtra)  # let R Markdown generate a LaTeX-based PDF
library(olsrr)       # package for p-value backward elimination
library(MASS)        # Functions & datasets for Venables & Ripley, "Modern Applied Statistics with S"
library(rlang)
library(gt)
```

Importing and Organizing Data

```
df  <- read_csv('../data/salesSFapr2023toMarch2025Updated.csv') |>
  mutate(across(where(is.labelled), as_factor)) |>
  rename_with(toupper)
```

Custom Functions

```
ftable <- function(data, variable, digits = 1) {
# data : data frame | variable : unquoted column name | digits : num decimals for the percent
  tbl <- data %>%                                # 1) build the basic tabyl, add % formatting and totals
    tabyl({{ variable }}) %>%                    # frequency table
    adorn_pct_formatting(digits = digits) %>%    # "12.3%"
    adorn_totals()                               # add a "Total" column
  mat <- t(tbl)                                  # 2) transpose to get categories across the top
  colnames(mat) <- rep("", ncol(mat))            # 3) blank out the auto-generated 1,2,3… column names
  print(mat)                                     # return as a matrix, or wrap in kable() downstream
}

graphing <- function(data, x, y, color, xlab, ylab, title) {
  xq <- enquo(x)
  yq <- enquo(y)
  ggplot(data, aes(x = !!xq, y = !!yq)) +
    geom_point() +
    stat_smooth(method = "lm", se = FALSE, color = color) +
    stat_regline_equation(
      aes(label =  paste(after_stat(eq.label), after_stat(adj.rr.label), sep = "~~~")),
      label.x.npc = 0.2,     # use "npc" coordinates 0-1 in panel
      label.y.npc = 0.8
    ) +
    labs(x = xlab, y = ylab, title = title) +
    theme_minimal()
}
```

Syntax Reference File - templateSyntaxECFs-aug25.sps

Initial Data Import Check

```
ftable(df, SALEPRICE, digits = 1)     # Frequencies of SALEPRICE | Line 23
psych::describe(df$SALEPRICE)
```

```
#df %>% distinct(SRESB_FIREPLACE)
df |>                                         # CROSSTABS Sresb_style BY Spropclass | Line 71
  tabyl(SRESB_STYHGT, SRESB_STYLE) |> adorn_totals(c("row","col")) |>
  adorn_percentages("row") |> adorn_pct_formatting(digits = 1) |> adorn_ns(position = "front")
```

```
graphing(df, STOTALSQFT, SALEPRICE, 'cyan',
         'Months', 'Time Adjusted Sales Price', 'TASP By Month')
```

Organize Data I

```
#    1. Price Categories
df <- df |>
  mutate(
    PRICECLASS = case_when(    # Lines 25 -> 34
      SALEPRICE < 48000                          ~ 1,
      SALEPRICE < 67000  & SALEPRICE >= 48000  ~ 2,
      SALEPRICE < 94000  & SALEPRICE >= 67000  ~ 3,
      SALEPRICE < 134000 & SALEPRICE >= 94000  ~ 4,
```

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      SALEPRICE >= 134000                      ~ 5
    ),
    PRICECLASS = factor(       # Lines 36 -> 41
      PRICECLASS,
      levels = 1:5,
      labels = c(
        "Below 48000", "48000 to 67000", "67000 to 94000", "94000 to 134000", "134000+"
      )
    )
  )

df <- df |>
  #   3. Filter to Relevant Styles    Lines 60 -> 67
  filter(SRESB_STYLE %in% c("SINGLE FAMILY", "1/2 DUPLEX", "ROW HOUSE")) |>
  #   4. Deduplicate Records    Lines 93 -> 122
  arrange(PARCELNO, SALEPRICE, SALEDATE) |>
  group_by(PARCELNO, SALEPRICE, SALEDATE) |>
  mutate(
    PRIMARYFIRST = row_number() == 1,
    PRIMARYLAST  = row_number() == n()
  ) |>
  ungroup() |>
  filter(PRIMARYLAST) |>
  #   5. Exclude Assessment-Equal Sales    Lines 129 -> 140
  mutate(
    PRICEISASMT_RATIO  = AVWHENSOLD / SALEPRICE,
    PRICEISASMT_RATIO2 = if_else(between(PRICEISASMT_RATIO, .90, 1.10), 1, PRICEISASMT_RATIO),
    PRICEISASMT        = if_else(PRICEISASMT_RATIO == 1, 0L, 1L)
  ) |>
  filter(PRICEISASMT == 1) |>
  #   6. Outlier Removal    Lines 71 -> 87
  mutate(
    OUT = case_when(
      STOTALSQFT      < 400   ~ 1L,
      SRESB_FLOORAREA < 400   ~ 1L,
      SALEPRICE       < 10000 ~ 1L,
      TRUE                    ~ 0L
    )
  ) |>
  filter(OUT == 0) |>
  #   8. Story-Style Dummies    9. Year-Built Era  Lines 225 -> 330
  mutate(
    ONESTORY            = as.integer(SRESB_STYHGT <= 2),
    TWOSTORY            = as.integer(SRESB_STYHGT > 2 & SRESB_STYHGT <= 5),
    SINGLE_FAMILY1STORY = as.integer(SRESB_STYLE == "SINGLE FAMILY" & ONESTORY == 1),
    SINGLE_FAMILY2STORY = as.integer(SRESB_STYLE == "SINGLE FAMILY" & TWOSTORY == 1),
    HALF_DUPLEX1STORY   = as.integer(SRESB_STYLE == "1/2 DUPLEX" & ONESTORY == 1),
    HALF_DUPLEX2STORY   = as.integer(SRESB_STYLE == "1/2 DUPLEX" & TWOSTORY == 1),
    ERABUILT = case_when(
      SRESB_YEARBUILT <  1910                           ~ 1,
      SRESB_YEARBUILT <  1930 & SRESB_YEARBUILT >= 1910 ~ 2,
      SRESB_YEARBUILT <  1946 & SRESB_YEARBUILT >= 1930 ~ 3,
      SRESB_YEARBUILT <  1961 & SRESB_YEARBUILT >= 1946 ~ 4,
      SRESB_YEARBUILT <= 1990 & SRESB_YEARBUILT >= 1961 ~ 5,
      SRESB_YEARBUILT >  1990                           ~ 6
    ),
    YB_PRE_1929     = as.integer(ERABUILT %in% c(1,2)), YB_1930_TO_1945 = as.integer(ERABUILT == 3),
    YB_1946_TO_1960 = as.integer(ERABUILT == 4),        YB_POST_1961    = as.integer(ERABUILT %in% c(5,6))
```

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) |>
#    10. Condition Dummies    Lines 334 -> 386
mutate(
  EXCELLENT          = as.integer(SCOND == 0),
  VERY_GOOD          = as.integer(SCOND == 1),
  GOOD               = as.integer(SCOND == 2),
  AVERAGE            = as.integer(SCOND == 3),
  FAIR               = as.integer(SCOND == 4),
  POOR               = as.integer(SCOND == 5),
  VERY_POOR_UNSOUND  = as.integer(SCOND %in% c(6,7)),
  TWO_PLUS_BATHS = as.integer(SRESB_FULLBATHS >= 2), TWO_BATHS = as.integer(SRESB_FULLBATHS == 2),
  POWDER_ROOM    = as.integer(SRESB_HALFBATHS >= 1), ONE_BATH  = as.integer(SRESB_FULLBATHS == 1),
  TOTAL_BATHS    = as.integer(SRESB_FULLBATHS + 0.5 * SRESB_HALFBATHS),
  FIREPLACE      = as.integer(SRESB_FIREPLACE >= 1)
) |>
#    12. Basement & Garage    Lines 393 -> 473
mutate(
  PCTCRAWL       = SRESB_CRAWSPACE / SRESB_GROUNDAREA,
  PCTSLAB        = SRESB_SLABAREA / SRESB_GROUNDAREA,
  PCTBSMNT       = SRESB_BASEMENTAREA / SRESB_GROUNDAREA,
  CRAWL          = as.integer(PCTCRAWL > PCTBSMNT & PCTCRAWL > PCTSLAB),
  SLAB           = as.integer(PCTSLAB  > PCTCRAWL & PCTSLAB  > PCTBSMNT),
  GARAGE         = as.integer(SRESB_GARAGEAREA >= 200),
  ONE_CAR_GARAGE = as.integer(SREB_GARTYPE == 1),
  TWO_CAR_GARAGE = as.integer(SREB_GARTYPE == 2),
  GARAGESPACES   = case_when(
    SRESB_GARAGEAREA > 200 & SRESB_GARAGEAREA < 420 ~ 1,
    SRESB_GARAGEAREA > 419 & SRESB_GARAGEAREA < 600 ~ 2,
    SRESB_GARAGEAREA > 599 & SRESB_GARAGEAREA < 820 ~ 3,
    SRESB_GARAGEAREA > 819                          ~ 4
  ),
  LN_GARAGESPACES = log(GARAGESPACES)
) |>
#    13. Quality & Street Class    Lines 478 -> 544
mutate(
  QABOVE_AVG      = as.integer(SRESB_BLDGCLASS %in% c(3,4,5)),
  Q_AVG           = as.integer(SRESB_BLDGCLASS == 2),
  QBELOW_AVG      = as.integer(SRESB_BLDGCLASS %in% c(0,1)),
  ARTERIAL        = as.integer(RD_CLASS == "A31"),
  FORCED_AIR      = as.integer(SRESB_HEAT %in% c(0,1)),
  HOT_WATER       = as.integer(SRESB_HEAT == 2),
  ELEC_BASEBOARD  = as.integer(SRESB_HEAT == 3),
  ELEC_RADIANT    = as.integer(SRESB_HEAT == 4),
  RADIANT_FLOOR   = as.integer(SRESB_HEAT == 5),
  ELEC_WALL       = as.integer(SRESB_HEAT == 6),
  SPACE_HEATER    = as.integer(SRESB_HEAT == 7),
  WALL            = as.integer(SRESB_HEAT == 8),
  CENTRAL_AIR     = as.integer(SRESB_HEAT == 9),
  SIZE = case_when(
    SRESB_FLOORAREA <   830                           ~ 1,
    SRESB_FLOORAREA <   1018 & SRESB_FLOORAREA >= 830  ~ 2,
    SRESB_FLOORAREA <   1266 & SRESB_FLOORAREA >= 1018 ~ 3,
    SRESB_FLOORAREA <   1659 & SRESB_FLOORAREA >= 1266 ~ 4,
    SRESB_FLOORAREA >=  1659                           ~ 5
  ),
  SMALLEST = as.integer(SIZE == 1), SMALL = as.integer(SIZE == 2),
  AVG = as.integer(SIZE == 3),      LARGE = as.integer(SIZE == 4),
  LARGEST = as.integer(SIZE == 5)
```

```r
) |>
#   14. Building Size & Lot Ratios    Lines 551 -> 593
  mutate(
    RATIO                    = SESTTCV / SALEPRICE,
    SPPSF                    = SALEPRICE / SRESB_FLOORAREA,
    BASE_BLDGSIZE            = case_when(
                               SRESB_STYLE == "SINGLE FAMILY" ~ 1402,
                               SRESB_STYLE == "1/2 DUPLEX"    ~ 860,
                               TRUE                           ~ NA_real_
                             ),
    BASE_BLDGSIZE_RATIO      = SRESB_FLOORAREA / BASE_BLDGSIZE,
    LN_BASE_BLDGSIZE_RATIO = log(BASE_BLDGSIZE_RATIO),
    LANDRATIO                = STOTALSQFT / case_when(
                               SRESB_STYLE == "SINGLE FAMILY" ~ 5009,
                               SRESB_STYLE == "1/2 DUPLEX"    ~ 3158.5,
                               TRUE                         ~ NA_real_
                             ),
    LANDRATIO2               = pmin(LANDRATIO, if_else(SRESB_STYLE == "SINGLE FAMILY",4,3)),
    LN_LANDRATIO             = log(LANDRATIO2),
    SALEDATE      = as_date(SALEDATE),
    MONTHS        = interval(min(SALEDATE), SALEDATE) %/% months(1), # LINE 498
    MONTHS_1TO9   = case_when(MONTHS <= 9  ~ MONTHS, TRUE ~ 9),
    MONTHS_10TO16 = case_when(MONTHS <= 9  ~ 0, MONTHS <= 16 ~ MONTHS - 9, TRUE ~ 7),
    MONTHS_17TO24 = case_when(MONTHS <= 16 ~ 0, TRUE ~ MONTHS - 16),
    LN_SPPSF      = log(SPPSF),
    LN_PRICE      = log(SALEPRICE)
  )
```

Organize Data II

```r
RATE1 <- -0.042539; RATE2 <- 0.195219; RATE3 <- 0.0; END_INDEX <- 1.3246

df <- df |>
#   15. Time Variables & Price Adjustment    Lines 520 -> 532
  mutate(
    PRICE_INDEX   = (1 + RATE1)^MONTHS_1TO9 * (1 + RATE2)^MONTHS_10TO16 * (1 + RATE3)^MONTHS_17TO24,
    TAF           = END_INDEX / PRICE_INDEX,
    TASP          = SALEPRICE * TAF,
    TASPPSF       = TASP / SRESB_FLOORAREA
  )
```